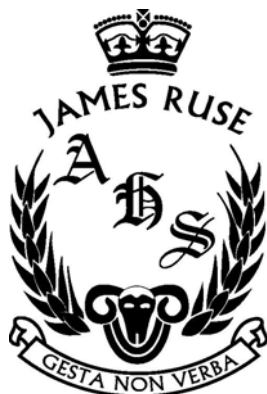


Name:	
Class:	



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2014 MATHEMATICS EXTENSION 1

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 2 hours.
- Write in black or blue pen.
- Board approved calculators & templates may be used
- A Standard Integral Sheet is provided.
- In Question 11 - 16, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 70

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 60 Marks

- Attempt Question 11 - 14
- Answer on blank paper unless otherwise instructed. Start a new page for each new question.
- Allow about 1 hours & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 11, Question 12, etc. Each question must show your Candidate Number.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice sheet for Questions 1-10

1. The focus of the parabola $(x - 3)^2 = -8y$ is
- a) $(3, -2)$ b) $(3, 2)$ c) $(0, -2)$ d) $(-2, 3)$
2. Find the acute angle between the lines $-x + 2y + 4 = 0$ and $3x + y + 1 = 0$. Give your answer to the nearest minute.
- a) $81^\circ 52'$ b) $78^\circ 41'$ c) $54^\circ 28'$ d) 45°
3. Find the coordinates of the point that divides the interval A $(-2, 7)$ B $(12, 0)$ externally in the ratio 4:3.
- a) $(-44, -28)$ b) $(44, -28)$ c) $(54, -21)$ d) $(54, 21)$
4. $\sin 2x$ equals to
- a) $\frac{1 - \tan^2 x}{1 + \tan^2 x}$ b) $\frac{2 \tan x}{1 + \tan^2 x}$ c) $\frac{2 \tan x}{1 - \tan^2 x}$ d) $\frac{1 + \tan^2 x}{1 - \tan^2 x}$
5. 8 people are to be seated around a circular table. If 2 people wish to sit next to each other, how many different ways can this be done?
- a) 720 b) 10080 c) 1440 d) 5040

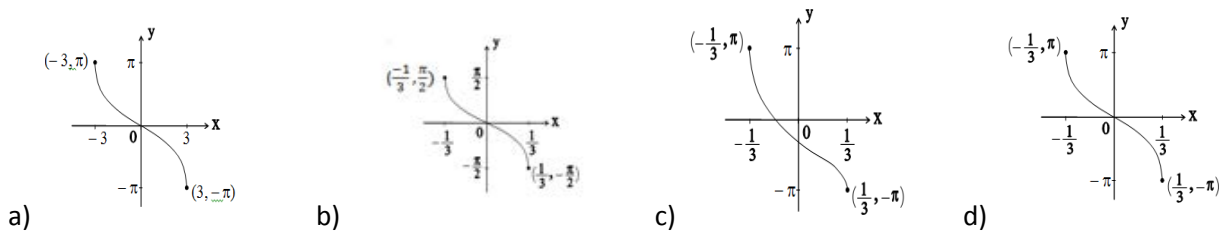
6. Use the trapezoidal rule to find an approximation for $\int_1^3 \log_e x \, dx$ using 2 subintervals.

- a) 1.09 b) 1.10 c) 1.24 d) 1.25

7. Find the $\lim_{x \rightarrow 0} \frac{\sin x \cos x}{2x}$

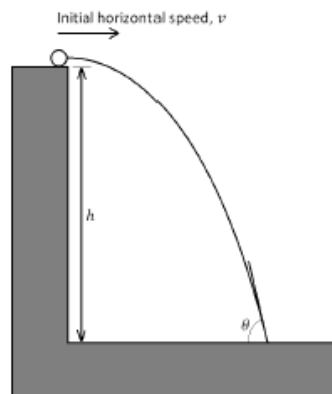
- a) 2 b) 1 c) $\frac{1}{2}$ d) $\frac{1}{4}$

8. Which of the following is the graph of $y = -2\sin^{-1}3x$?



9. The diagram below shows the path of a projectile fired with a horizontal velocity v from a cliff of height h .

Which pair of the following values of v and h will give the greatest value of angle θ ?



- a) $v = 10\text{ms}^{-1}$
 $h = 30\text{m}$ b) $v = 30\text{ms}^{-1}$
 $h = 50\text{m}$ c) $v = 50\text{ms}^{-1}$
 $h = 10\text{m}$ d) $v = 10\text{ms}^{-1}$
 $h = 50\text{m}$

10. The solution to

$$\frac{2t+1}{t-2} > 1$$

is

- a) $t > -3$ b) $t > 2 \text{ or } t < -3$ c) $t > -1$ d) $t > 3 \text{ or } t < -2$

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE page. Extra writing papers are available.

In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new page.

- a) Find 3

$$\int x\sqrt{(x^2 + 1)^3} \, dx$$

using the substitution $u = x^2 + 1$

- b) The area bounded by the curve $y = \cos 2x$, the x-axis and lines $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ is 3
rotated about the x-axis. Find the volume of the solid generated in exact form.

- c) There are 4 families, and each family has exactly 4 children. Assume that the probability 3
of giving birth to a male and giving birth to a female are even.

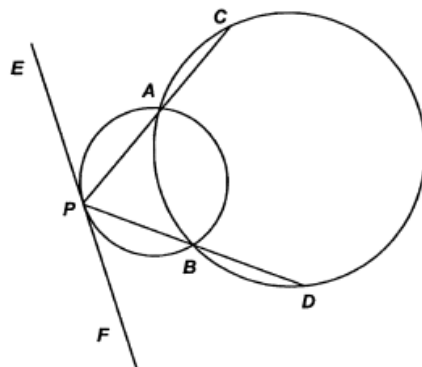
Determine the probability that exactly 2 of the families will have exactly 2 males and 2 females as children.

Question 11 continues on page 5

d)

Copy the diagram into your answer booklet. PAC , PBD are straight lines. EF is the tangent at P .

3



Prove $CD \parallel EF$

e)

Prove by mathematical induction that for any positive integer $n \geq 1$.

3

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$$

End of Question 11

Question 12 (15 marks) Start a new page.

a)

A body is in Simple Harmonic Motion and its position at a time t is given by the equation

$$x = R \cos(nt + \alpha) + 1$$

The period of motion is π seconds and $0 \leq \alpha \leq \frac{\pi}{2}$. Initially the body is at rest 3 units to the left of the origin.

- i. Find the values of R , n and α . 3
- ii. Find the velocity of the body when $t = \frac{\pi}{6}$ 1

b)

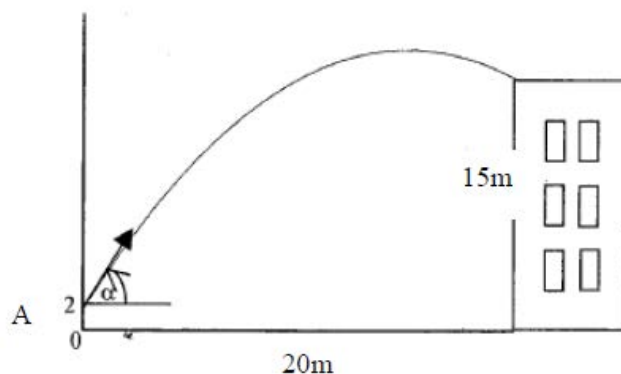
Show that the sum of the Arithmetic sequence 2

$$\log_{10}(x-2), \log_{10}(x-2)^2, \log_{10}(x-2)^3, \dots, \log_{10}(x-2)^n$$

is $\frac{n}{2} \log_{10}(x-2)^{n+1}$

c)

Andrew, whose height is 2 metres, throws a ball from area A in the direction of the Cohen building which is 15 metres high. He throws the ball with an initial velocity u at angle α , and he is 20 metres from the base of the building. (Assume $\ddot{x} = 0$ and $\ddot{y} = -10\text{m/s}^2$)



- i. Show that $y = x \tan \alpha - \frac{5x^2}{u^2} (1 + \tan^2 \alpha) + 2$, at any time t . 2
- ii. Hence, find between which two angles of projection must he throw the ball to ensure that it lands on the roof of the building, or over, given that $u = 25\text{m/s}$. (Answer to the nearest degrees). 3

d)

- i. Differentiate $x \tan^{-1} x$ 2
- ii. Hence, or otherwise, find $\int \tan^{-1} x \, dx$ 2

End of Question 12

Question 13 (15 marks) Start a new page.

a)

At any time t minutes, the rate of cooling of a body with temperature T , when the surrounding temperature is S , is given by the differential equation

$$\frac{dT}{dt} = -k(T - S)$$

for some constant k .

i. Show that $T = S + Ae^{-kt}$, for some constant A , satisfies this differential equation. **1**

ii. A metal rod has an initial temperature of 1390°C and cools to 1060°C in 10 minutes when the surrounding temperature is 30°C . **4**

Find how much longer it will take the rod to cool to 110°C , giving your answer to the nearest minute.

b)

Find the monic cubic equation whose roots are the squares of the roots of **3**

$$x^3 + 2x + 1 = 0$$

c)

The acceleration of a particle P is given by the equation

$$\ddot{x} = 18x^3 + 27x^2 + 9x$$

Initially $x = -2$ and the velocity, $v = -6$.

i. Show that $v^2 = 9x^2(1 + x)^2$ **2**

ii. Hence, or otherwise, show that **2**

$$\int \frac{1}{x(1+x)} dx = -3t + C, \text{ for some constant } C$$

iii. Find the derivative of $\log_e(1 + \frac{1}{x})$ **1**

iv. Using your result in part iii and the initial conditions, find x as a function of t . **2**

End of Question 13

Question 14 (15 marks) Start a new page.

a)

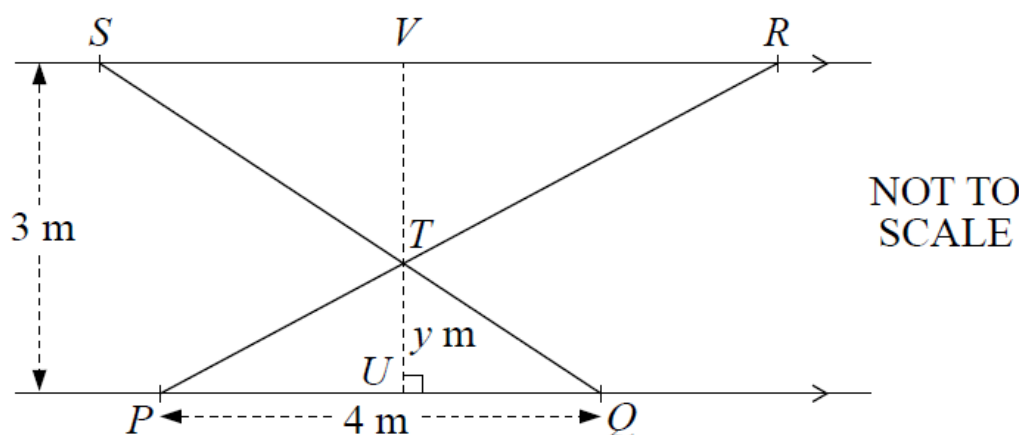
Show that $x = 5\sin\theta$ and $y = 5\cos\theta + 1$ satisfies the equation

2

$$y^2 + x^2 - 2y - 24 = 0$$

b)

In the diagram, PQ and SR are parallel railings which are 3 metres apart. The points P and Q are fixed 4 metres apart on the lower railing. Two crossbars PR and QS intersect at T as shown in the diagram. The line through T perpendicular to PQ intersects PQ at U and SR at V. The length of UT is y metres.



i. Deduce that $SR = \frac{12}{y} - 4$

3

ii. Hence show that the total area A of $\triangle PTQ$ and $\triangle RTS$ is

1

$$A = 4y - 12 + \frac{18}{y}$$

iii. Find the value of y that minimises A . Justify your answer.

3

c)

The coefficient of x^k in $(1 + x)^n$, where n is a positive integer, is denoted by c_k (so $c_k = {}^nC_k$)

i. Show that $c_0 + 2c_1 + 3c_2 + \dots + (n + 1)c_n = (n + 2)2^{n-1}$

3

ii. Find the sum,

3

$$\frac{c_0}{1 \times 2} - \frac{c_1}{2 \times 3} + \frac{c_2}{3 \times 4} - \dots + (-1)^n \frac{c_n}{(n + 1)(n + 2)}$$

Showing all necessary working.

End of paper



Extension One Trial

Student Number

2014

Q1. A

2. A

3. C

4. B

5. C

6. ~~B~~ C

7. C

8. D

9. D

10. B

MATHEMATICS Extension 1 : Question...

Suggested Solutions	Marks	Marker's Comments
a) Let $I = \int x \sqrt{x^2+1}^3 dx$ let $u = x^2+1$ $\frac{du}{dx} = 2x$ $\therefore du = 2x dx$ $I = \int \frac{1}{2} \sqrt{u}^3 du$ $= \frac{1}{2} \left[\frac{u^{5/2}}{5/2} \right]$ $= \frac{u^{5/2}}{5}$ $= \frac{(x^2+1)^{5/2}}{5}$	1	
b) $V = \pi \int_{\pi/4}^{3\pi/4} (\cos 2x)^2 dx$ $= \pi \int_{\pi/4}^{3\pi/4} \cos^2 2x dx$	1	Many students failed to substitute back Some 2 students made this question harder by trying to split it up.

MATHEMATICS Extension 1: Question....11.

Suggested Solutions

Marks

Marker's Comments

$$= \frac{\pi}{2} \int_{\pi/4}^{3\pi/4} \cos 2x + 1 dx$$

$$= \frac{\pi}{2} \left[\frac{1}{4} \sin 4x + x \right]_{\pi/4}^{3\pi/4}$$

$$= \frac{\pi}{2} \left[\left[\frac{1}{4} 0 + 3\pi/4 \right] - \left[0 + \pi/4 \right] \right]$$

$$= \pi/2 \times \pi/2$$

$$= \frac{\pi^2}{4} \text{ units}^3$$

Students who did not integrate $\int \cos 2x dx$ made question easier so could not gain carry forward errors.

Some careless calculation errors.

c) Probability of exactly 2 males and 2 females = ${}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2$

$$= \frac{3}{8}$$

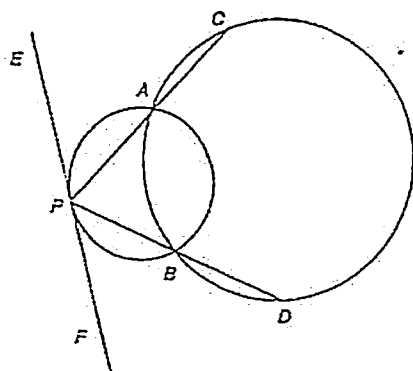
Pr(exactly 2 families have exactly 2 males and 2 females)

$$= {}^4C_2 \left(\frac{3}{8}\right)^2 \left(\frac{5}{8}\right)^2$$

$$= \frac{675}{2048}$$

many students stopped here.

1 for 4C_2 missing
4 for common
1 for $\left(\frac{3}{8}\right)^2 \left(\frac{5}{8}\right)^2$ error



Construct AB and CD

let $\angle EPA = \alpha$

$\angle ABP = \alpha$ (angle between tangent and chord equals angle at circumference in alternate segment)

$\angle ACD = \alpha$ (exterior angle of cyclic quadrilateral ABCD equals opposite interior angle)

$\therefore \angle EPA = \angle ACD = \alpha$

$\therefore CD \parallel EF$ (alternate angles are equal)

students used opposite rather than alternate which is incorrect.

MATHEMATICS Extension 1 : Question...\\...

Suggested Solutions

Marks

Marker's Comments

e) RTP

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = n$$

Prove true for $n=1$

$$\text{LHS } \frac{1}{1 \times 5} = \frac{1}{5}$$

$$\text{RHS } \frac{1}{4+1} = \frac{1}{5}$$

$\therefore$ true for $n=1$

Assume true for $n=k$. $k \in \mathbb{Z}^+$

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$$

Prove true for $n=k+1$

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4k+1)(4k+5)} = \frac{k+1}{4k+5}$$

by assumption

$$\begin{aligned} \text{LHS} &= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)} \\ &= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)} \end{aligned}$$

Must clearly
make substitution
into LHS and
RHS

1
for
scaffolding

MATHEMATICS Extension 2: Question 11....

Suggested Solutions

Marks

Marker's Comments

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}$$

$$= \frac{k+1}{4k+5}$$

By principle of mathematical induction true for all positive integers $n \geq 1$

1

MATHEMATICS Extension 1 : Question 12

Suggested Solutions

Marks

Marker's Comments

$$a) x = R \cos(n\pi + \alpha) + 1$$

period is π

$$\frac{2\pi}{n} = \pi$$

$$\therefore n = 2$$

$$\dot{x} = -\cancel{R\sin} - 2R \sin(2\pi + \alpha)$$

$$\text{when } t = 0 \quad \dot{x} = 0$$

$$\therefore 0 = -2R \sin \alpha$$

$$\text{Since } 0 \leq \alpha \leq \pi/2$$

$$\alpha = 0$$

$$x = R \cos 2t + 1$$

$$\text{when } x = -3 \quad t = 0$$

$$-3 = R + 1$$

$$\therefore R = -4$$

1

Students who did not calculate α first had trouble with R.

1

1

students changed to -4 stating $R > 0$

$$b) a = \log_{10}(x-2)$$

$$d = \log_{10}(x-2)$$

$$\text{last term } \log_{10}(x-2)^n$$

n terms

$$S_n = \frac{n}{2} (a + d)$$

$$= \frac{n}{2} (\log_{10}(x-2) + \log_{10}(x-2)^n)$$

$$= \frac{n}{2} (\log_{10}(x-2)^{n+1})$$

$$\text{also } \frac{n}{2} (2a + (n-1)d) \text{ also worked}$$

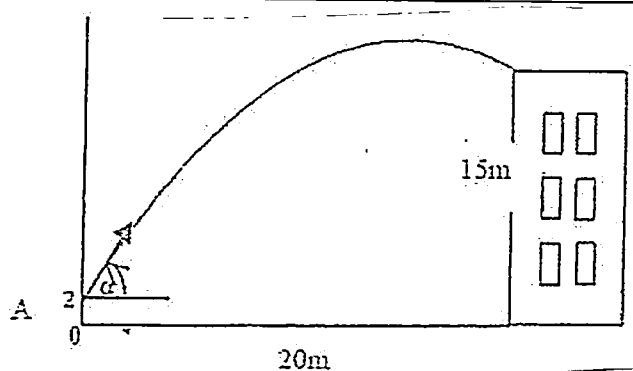
Q

Ex+1 MATHEMATICS: Question...12...

Suggested Solutions

Marks

Marker's Comments



i) $\ddot{y} = -10$

$\ddot{x} = 0$

$\dot{y} = -10t + c_1$

$\dot{x} = c_3$

$t=0 \quad \dot{y} = u \sin \alpha$

when $c_3 = u \cos \alpha \quad t=0$

$\dot{x} = u \cos \alpha$

∴ $c_1 = u \sin \alpha$

$x = ut \cos \alpha + c_4$

$\dot{y} = -10t + u \sin \alpha$

when $t=0 \quad x=0$

$\dot{y} = -5t^2 + ut \sin \alpha + c_2$

$x = ut \cos \alpha$

$t=0 \quad y=2$

∴ $2 = c + c_2$

$t = \frac{x}{u \cos \alpha}$

$c_2 = 2$

$y = -5t^2 + ut \sin \alpha + 2$

$y = -5 \left(\frac{x}{u \cos \alpha} \right)^2 + u \times \frac{x}{\cos \alpha} \sin \alpha + 2$

$= -\frac{5x^2}{u^2 \cos^2 \alpha} + x \tan \alpha + 2$

$= -\frac{5x^2}{u^2} \sec^2 \alpha + x \tan \alpha + 2$

$= x \tan \alpha - \frac{5x^2}{u^2} (1 + \tan^2 \alpha) + 2$

since $\sec^2 \alpha = 1 + \tan^2 \alpha$

many students
failed to
include
u-sin-alpha

needed to
have sec^2 alpha
somewhere in
solution

Ext 1 MATHEMATICS: Question.....

Suggested Solutions

Marks

Marker's Comments

$$y \geq 15 \quad u = 25 \text{ m/s} \quad x = 20$$

$$15 \leq 20 + \frac{-5 \times (20)^2}{25^2} - \frac{5 \times 20^2}{25^2} \tan^2 \alpha + 2$$

$$\therefore 16 \tan^2 \alpha - 100 \tan \alpha + 81 \leq 0$$

$$\text{for } 16 \tan^2 \alpha - 100 \tan \alpha + 81 = 0$$

$$\tan \alpha = \frac{100 \pm \sqrt{100^2 - 4 \times 81 \times 16}}{32}$$

$$= \frac{25 \pm \sqrt{301}}{8}$$

$$\tan \alpha = 5.2936 \text{ or } 0.956331$$

$$\therefore 44^\circ \leq \alpha \leq 79^\circ$$

$$d(u) \quad \frac{dy}{dx} = \frac{1}{x} \tan^{-1} x + \frac{x}{1+x^2}$$

$$(u) \quad \int \tan^{-1} x \, dx = \int \frac{d}{dx} x \tan^{-1} x \, dx - \int \frac{x}{1+x^2} \, dx$$

$$\therefore \int \tan^{-1} x \, dx = x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$$

Some students used 13 but equation had already taken 2 into account.

Question well done.

1 for $x \tan^{-1} x + C$
1 for $-\frac{1}{2} \ln(1+x^2)$

2014 TRIAL MATHEMATICS Extension 1 : Question 14...

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned}
 i) \quad y^2 + x^2 - 2y - 24 &= 0 \\
 &= (5\cos\theta + 1)^2 + (5\sin\theta)^2 - 2(5\cos\theta + 1) - 24 \\
 &= 25\cos^2\theta + 10\cos\theta + 1 + 25\sin^2\theta - 10\cos\theta - 2 - 24 \\
 &= 25(\cos^2\theta + \sin^2\theta) - 25 \\
 &= 25(1) - 25 \quad (\text{as } \sin^2\theta + \cos^2\theta = 1) \\
 &= 0 \\
 &= \text{RHS.}
 \end{aligned}$$

①

①

ii) i) In $\triangle STR$, PTQ $\angle STR = \angle PTQ$ (vertically opposite angles are equal) $\angle TSR = \angle TQP$ (alternate angles are equal, $SR \parallel QP$) $\therefore \triangle STR \parallel \triangle PTQ$ (equiangular) $\frac{SR}{PQ} = \frac{ST}{PT}$ (corresponding sides in similar triangles are in the same ratio)

$$\begin{aligned}
 SR &= 4 \times \frac{(3-y)}{y} \\
 &= \frac{12}{y} - 4
 \end{aligned}$$

①

* lost a mark if you didn't write equiangular.

ii) Total area = $\triangle PTQ + \triangle STR$

$$= \frac{1}{2} \times 4 \times y + \frac{1}{2} \times \left(\frac{12}{y} - 4\right)(3-y)$$

$$= 2y + \left(\frac{6}{y} - 2\right)(3-y)$$

$$= 2y + \frac{18}{y} - 6 - 6 + 2y$$

$$= 4y - 12 + \frac{18}{y}$$

①

$$iii) A = 4y - 12 + \frac{18}{y}$$

$$\frac{dA}{dy} = 4 - \frac{18}{y^2}$$

$$\frac{d^2A}{dy^2} = \frac{36}{y^3}$$

MATHEMATICS Extension 1 : Question 14

Suggested Solutions	Marks	Marker's Comments
possible stat. pts when $\frac{dA}{dy} = 0$ ie $4 - \frac{18}{y^2} = 0$ $4y^2 = 18$ $y^2 = \frac{18}{4} = \frac{9}{2}$ $y = \pm \frac{3}{\sqrt{2}}$ $y > 0$ as it's a length; so $y = \frac{3}{\sqrt{2}}$ only when $y = \frac{3}{\sqrt{2}}$ $\frac{d^2A}{dy^2} = \frac{8}{3\sqrt{2}} > 0$ $\therefore$ concave up $\therefore$ relative minimum at $y = \frac{3}{\sqrt{2}}$ since there's only one stat. pt in the domain $y > 0$, the relative minimum is the absolute minimum.		① *lost 1mk if you "lost" a solution without explanation ① ①
Q1 (i) $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$ let $x=1$ $\therefore (1+1)^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$ $\therefore 2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$ --- ① differentiate wrt x the first eqn. $n(1+x)^{n-1} = {}^nC_1 + 2{}^nC_2x + 3{}^nC_3x^2 + \dots + n{}^nC_nx^{n-1}$ sub. into $x=1$ $n2^{n-1} = {}^nC_1 + 2{}^nC_2 + 3{}^nC_3 + \dots + n{}^nC_n$ --- ② ② + ① $2^n + n2^{n-1} = {}^nC_0 + {}^nC_1 + {}^nC_1 + {}^nC_2 + 2{}^nC_2 + {}^nC_3 + 3{}^nC_3 + \dots + {}^nC_n + n{}^nC_n$ $2^{n-1}(2+n) = {}^nC_0 + 2{}^nC_1 + 3{}^nC_2 + 4{}^nC_3 + \dots + (n+1){}^nC_n$		poorly set out If you didn't show the last term you lost a mark. ① ① ① ①

B/A

MATHEMATICS Extension 1 : Question 14...

Suggested Solutions

Marks

Marker's Comments

$$22(ii) (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

integrate both sides wrt x

$$\frac{(1+x)^{n+1}}{n+1} + K_1 = {}^nC_0 x + \frac{{}^nC_1 x^2}{2} + \frac{{}^nC_2 x^3}{3} + \dots + \frac{{}^nC_n x^{n+1}}{n+1}$$

sub in $x=0$

$$\frac{1}{n+1} + K_1 = 0$$

$$K_1 = -\frac{1}{n+1}$$

$$\therefore \frac{(1+x)^{n+1}}{n+1} - \frac{1}{n+1} = {}^nC_0 x + \frac{{}^nC_1 x^2}{2} + \frac{{}^nC_2 x^3}{3} + \dots + \frac{{}^nC_n x^{n+1}}{n+1}$$

Integrate both sides wrt x

$$\frac{(1+x)^{n+2}}{(n+1)(n+2)} - \frac{x}{n+1} + K_2 = \frac{{}^nC_0 x^2}{2} + \frac{{}^nC_1 x^3}{2 \times 3} + \frac{{}^nC_2 x^4}{3 \times 4} + \dots + \frac{{}^nC_n x^{n+2}}{(n+1)(n+2)}$$

sub in $x=0$

$$\frac{1}{(n+1)(n+2)} - 0 + K_2 = 0$$

$$K_2 = \frac{-1}{(n+1)(n+2)}$$

$$\therefore \frac{(1+x)^{n+2}}{(n+1)(n+2)} - \frac{x}{n+1} - \frac{1}{(n+1)(n+2)} = \frac{{}^nC_0 x^2}{1 \times 2} + \frac{{}^nC_1 x^3}{2 \times 3} + \dots + \frac{{}^nC_n x^{n+2}}{(n+1)(n+2)}$$

sub in $x=-1$

$$0 + \frac{1}{(n+1)} - \frac{1}{(n+1)(n+2)} = \frac{{}^nC_0}{1 \times 2} - \frac{{}^nC_1}{2 \times 3} + \frac{{}^nC_2}{3 \times 4} - \dots + \frac{(-1)^n {}^nC_n}{(n+1)(n+2)}$$

$$\frac{n+3}{(n+1)(n+2)} - \frac{1}{(n+1)(n+2)} = \frac{{}^nC_0}{1 \times 2} - \frac{{}^nC_1}{2 \times 3} + \frac{{}^nC_2}{3 \times 4} - \dots + \frac{(-1)^n {}^nC_n}{(n+1)(n+2)}$$

$$\text{series} = \frac{n+1}{(n+1)(n+2)} = \frac{1}{n+2}$$

(1)

(1)

(1)

MATHEMATICS Extension 1 : Question 14...

Suggested Solutions

Marks

Marker's Comments

$$\textcircled{1} (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

integrate both sides

$$\frac{(1+x)^{n+1}}{n+1} = {}^nC_0 x + \frac{{}^nC_1 x^2}{2} + \frac{{}^nC_2 x^3}{3} + \dots + \frac{{}^nC_n x^{n+1}}{n+1}$$

Integrate both sides

$$\frac{(1+x)^{n+2}}{(n+1)(n+2)} = \frac{{}^nC_0 x^2}{1 \times 2} + \frac{{}^nC_1 x^3}{2 \times 3} + \frac{{}^nC_2 x^4}{3 \times 4} + \dots + \frac{{}^nC_n x^{n+2}}{(n+1)(n+2)}$$

substitute $x = -1$

$$\frac{(1-1)^{n+2}}{(n+1)(n+2)} = +\frac{{}^nC_0}{1 \times 2} - \frac{{}^nC_1}{2 \times 3} + \frac{{}^nC_2}{3 \times 4} - \dots - \frac{(-1)^n {}^nC_n}{(n+1)(n+2)}$$

$$\therefore 0 = \frac{{}^nC_0}{1 \times 2} - \frac{{}^nC_1}{2 \times 3} + \frac{{}^nC_2}{3 \times 4} - \dots + \frac{(-1)^n {}^nC_n}{(n+1)(n+2)}$$

$\therefore$ sum of series is 0.

① for integrating twice + subbed in $x = -1$.

① for attempting to evaluate the constant of integration

① for fully correction solution

Suggested Solutions

Marks

Marker's Comments

a) i) Substitute $T = S + Ae^{-kt}$ into eqn.

$$\begin{aligned} \text{LHS} &= \frac{dT}{dt} \\ &= -kAe^{-kt} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= -k(T-S) \\ &= -k(Ae^{-kt}) \quad \text{from above} \\ &= -kAe^{-kt} \end{aligned}$$

$$\text{LHS} = \text{RHS}$$

So $T = S + Ae^{-kt}$ is a solution of

$$\frac{dT}{dt} = -k(T-S)$$

ii) $S = 30$ and when $t = 0$, $T = 1390$

$$\therefore 1390 = 30 + A$$

$$\underline{A = 1360}$$

When $t = 10$, $T = 1060$

$$\therefore 1060 = 30 + 1360e^{-10k}$$

$$e^{-10k} = \frac{1030}{1360}$$

$$10k = \ln\left(\frac{136}{103}\right)$$

$$\underline{k = \frac{1}{10} \ln\left(\frac{136}{103}\right)} \quad (= 0.02779)$$

Find t when $T = 110$

$$110 = 30 + 1360e^{-\frac{t}{10} \ln\left(\frac{136}{103}\right)}$$

$$\therefore \frac{1360}{80} = e^{\ln\left(\frac{136}{103}\right) \frac{t}{10}}$$

$$\frac{136}{8} = \left(\frac{136}{103}\right)^{t/10}$$

A show question needs a little more structure than one line.

Most people got these 2 marks

METHOD 2

α, β, γ are x values

If y has values $\alpha^2, \beta^2, \gamma^2$ then

$$y = x^2$$

$$x = \sqrt{y}$$

Substitute into original equation

$$y\sqrt{y} + 2\sqrt{y} + 1 = 0$$

$$\sqrt{y}(y+2) = -1$$

$$y(y+2)^2 = (-1)^2$$

$$y(y^2+4y+4) = 1$$

$$\underline{\underline{y^3 + 4y^2 + 4y - 1 = 0}}$$

1

1

1

c) i) $\ddot{x} = \frac{d}{dx} \left(\frac{v^2}{2} \right) = 18x^3 + 27x^2 + 9x$

$$\therefore -\frac{v^2}{2} = \frac{18x^4}{4} + 9x^3 + \frac{9x^2}{2} + k_1$$

$$v^2 = 9x^4 + 18x^3 + 9x^2 + k_2$$

$$= 9x^2(x^2 + 2x + 1) + k_2$$

$$= 9x^2(x+1)^2 + k_2$$

When $x = -2$, $v = -6$.

$$\therefore 36 = 9 \times 4 \times (-1)^2 + k_2$$

$$\therefore k_2 = 0$$

$$\therefore \underline{\underline{v_2 = 9x^2(x+1)^2}}$$

One mark for
integrating and
evaluating
constant.

One for tidying
into given form.

1

1

c) ii) $v^2 = 9x^2(1+x)^2$

$$\therefore v = \pm 3x(1+x)$$

Initially, $v = -6$ (moving to left)
and $x = -2$

and acceleration $= -54$ also to left.

$\therefore$ velocity always negative.

$$v = \frac{dx}{dt} = -3x(1+x)$$

$$\therefore \int \frac{dx}{x(1+x)} = -3 \int dt$$

$$\underline{\underline{\int \frac{dx}{x(1+x)} = -3t + C}}$$

iii) $\frac{d}{dx} \left(\ln \left(1 + \frac{1}{x} \right) \right) = \frac{-1/x^2}{1 + 1/x}$

$$= \frac{-1}{x^2 + x} = \underline{\underline{\frac{-1}{x(x+1)}}}$$

iv) From (iii) $\int \frac{dx}{x(1+x)} = -\ln \left(1 + \frac{1}{x} \right) + K$

From (ii) $-3t + C = -\ln \left(1 + \frac{1}{x} \right) + K$

$$3t = \ln \left(1 + \frac{1}{x} \right) + K_2$$

When $t = 0$, $x = -2 \therefore \underline{\underline{K_2 = \ln 2}}$

$$\therefore e^{3t} = 2 \left(1 + \frac{1}{x} \right)$$

$$e^{3t} - 2 = \frac{2}{x}$$

$$\therefore x = \underline{\underline{\frac{2}{e^{3t} - 2}}}$$

The first mark needed hard work if you didn't start with $\pm$ sign.

People should try to be consistent with constants. Don't use same letter if it has been manipulated.